Right-To-Left and Left-To-Right characters

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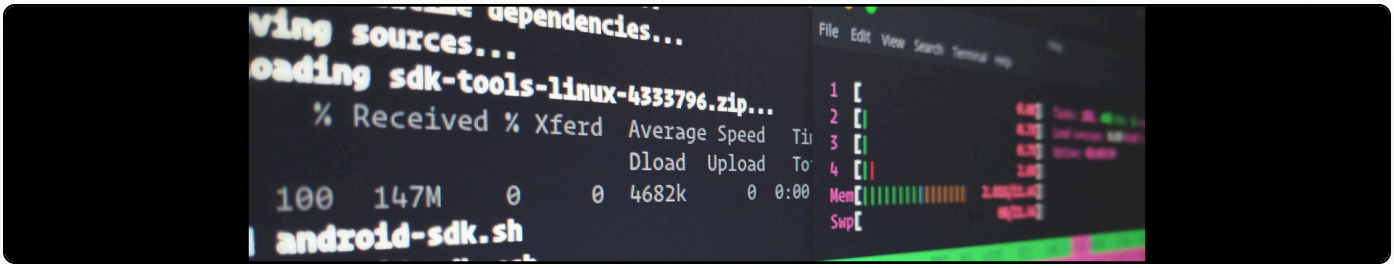

There's been a fair bit of discussion going on at slackers on the security implications of the Unicode characters U+202D and U+202E which switch the left-to-right and right-to-left encoding of the following text.

So, what you appear to have in the source is:

```
<html id="test">
<head><title>A Test</title></head>
<body>
[REVERSE CHAR]<script>alert(1)</script>[UNREVERSE CHAR]
</body>
</html>
```

Which instantly leads to the question: is that text reversed and could therefore this be used for filter evasion?

To investigate, I created a simple c# program that creates 2 strings, the only difference between them being the inclusion of the reverse characters.

```
string s = "\r\n";
s += (char)int.Parse("202E", System.Globalization.NumberStyles.HexNumber);
s += TextBox1.Text;
s += (char)int.Parse("202D", System.Globalization.NumberStyles.HexNumber) + "\r\n";

string s2 = "\r\n";
s2 += TextBox1.Text;
s2 += "\r\n";
```

When cast to a char array, the output looked like this:

> *String containing evil characters: 13, 10, 8238, 60, 115, 99, 114, 105, 112, 116, 62, 97, 108, 101, 114, 116, 40, 49, 41, 60, 47, 115, 99, 114, 105, 112, 116, 62, 8237, 13, 10*
>
> *String without: 13, 10, 60, 115, 99, 114, 105, 112, 116, 62, 97, 108, 101, 114, 116, 40, 49, 41, 60, 47, 115, 99, 114, 105, 112, 116, 62, 13, 10*

I'll save you the hassle of looking and tell you now that, under .NET anyway, they are exactly the same. This means that any regex matching or String.Contains() functions will return the correct value and these representations will not evade filters. Whether PHP does the same, I shall leave for someone else to discover.

More disturbing however is the fact that these characters appear to be ignored by browser parsers meaning that putting one halfway through a word could lead to potential filter evasion as the string is not left in tact.